

Substructuring Techniques in Structural Dynamics

A Comparative Study and Application to Aerospace Structures

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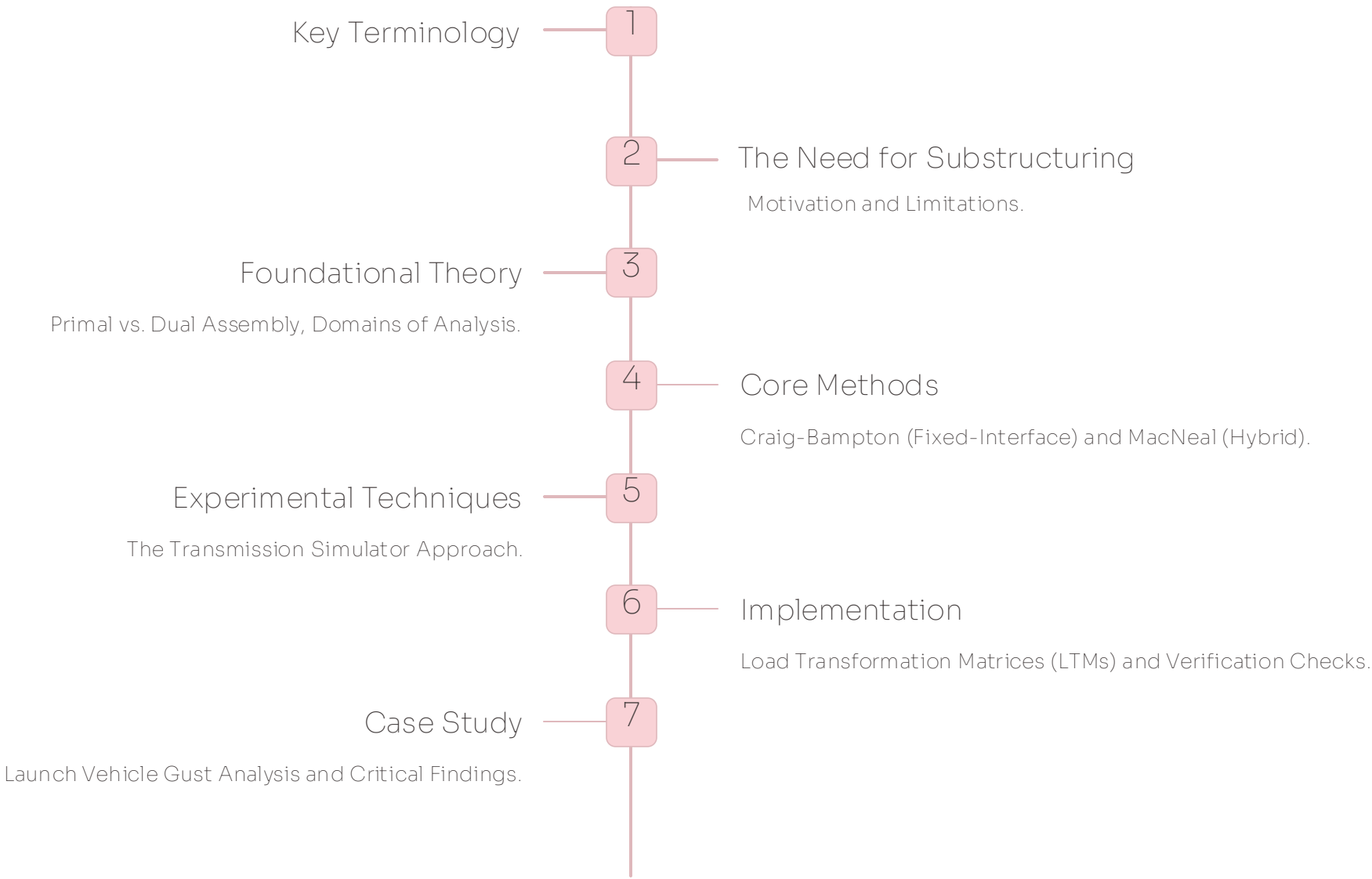
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Presentation Structure

This presentation follows a logical flow, starting with the theoretical necessity of Substructuring and reducing complexity in large Finite Element Models (FEMs), then exploring established and modern Reduction of Order Methods (ROMs), and concluding with verification and a practical application in launch vehicle gust analysis.



Key Terminology

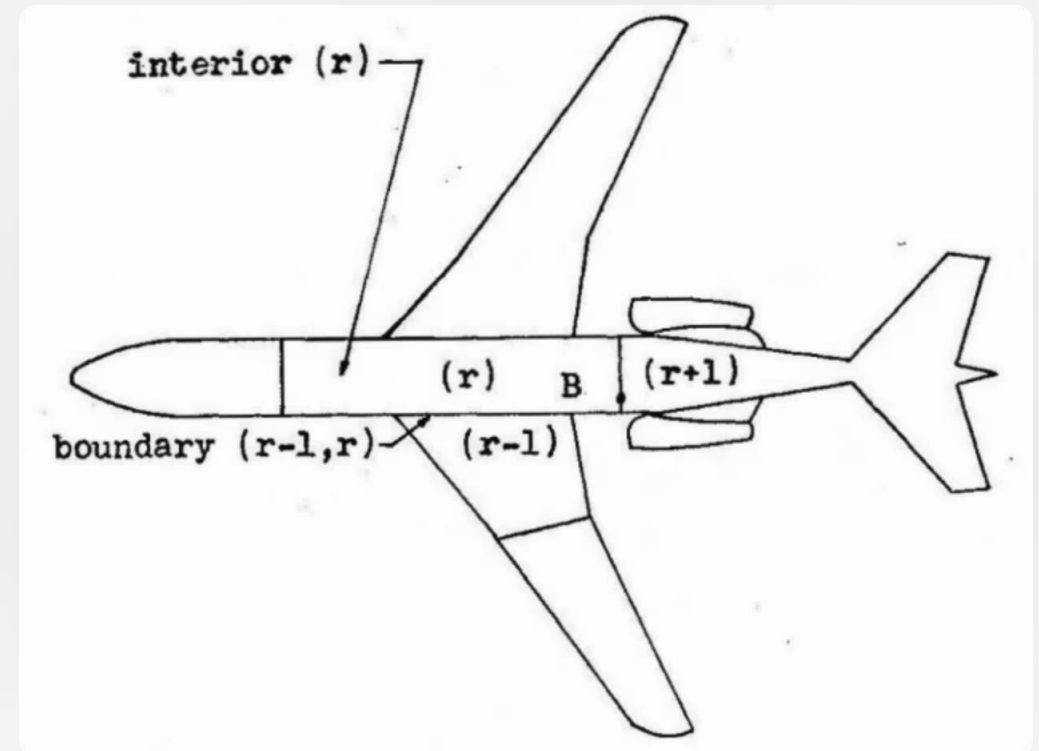
Glossary for Substructuring

Abbreviation	Full Name and Context
ROM	Reduction of Order Model: Any technique (like CMS) used to simplify massive Finite Element Models.
DOFs	Degrees of Freedom: The fundamental number of independent coordinates defining the state of a structure.
CMS	Component Mode Synthesis: The general class of ROM techniques based on modal analysis.
LTM / OTMs	Load / Observation Transformation Matrices: Post-processing tools to map results from the reduced modal space back into physical forces or displacements.
C-B	Craig-Bampton: The industry-standard fixed-interface method of CMS.
FRF	Frequency Response Function: Experimental data used to relate output response to input excitation across a frequency range.
TS	Transmission Simulator: Experimental methodology used to mathematically decouple the test structure from a compliant test fixture.
PHIZ	Modal Acceleration: A high-fidelity LTM technique used to recover accurate physical displacements from modal accelerations.

Motivation: Why Substructuring is Essential

The Challenge of Scale

Substructuring is the "divide and conquer" strategy for large aerospace models (millions of DOFs) where direct analysis is too slow. It allows companies to exchange condensed, proprietary models easily, reduces computation time for transient analysis, and efficiently isolates and solves local nonlinearities (like bolted joints).



Limitations of Direct FEM

Why Standard Methods Fall Short

Computational Burden

Full system integration in Finite Element Analysis (FEA) is simply too slow for effective, iterative design cycles.

Modal Truncation Issues

Pure modal truncation, where high-frequency modes are simply cut, results in models that are unrealistically stiff at the structural interfaces, leading to inaccurate load transfer predictions.

Manual Classification Problems

Older techniques, like those developed by Hurty, required time-consuming and error-prone manual classification of constraints, making them impractical for modern, highly indeterminate structures.

Testing Limitations

assumption of perfectly "fixed" (rigid) boundary conditions in modelling is often physically impossible to achieve during lab testing due to fixture compliance.

Domains of Dynamic Representation

Data Transformation for Efficiency

Structural dynamics data can be represented in three key domains, each serving a different analytical purpose.

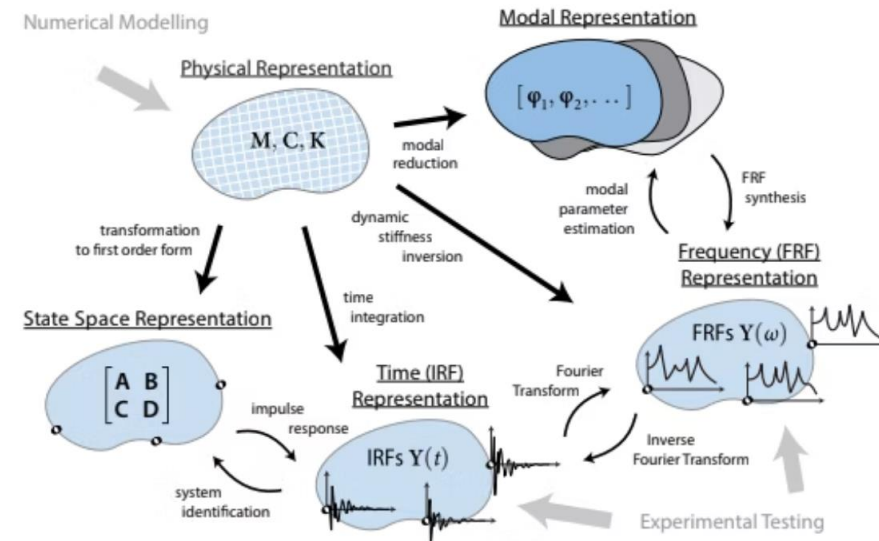


Figure 2: Schematic overview of substructuring domains and the transformations linking numerical modeling to experimental testing [1].

Physical Domain

Defined by Mass M , Stiffness K , and Damping C matrices.

The high-fidelity representation, but is computationally the most expensive.

Modal Domain

Projects the physical response onto a smaller subspace defined by basis vectors (eigenmodes).

Drastically reducing the system's order whilst preserving its essential dynamic characteristics.

Frequency Domain

Characterises the structure in experimental methods by relating the complex amplitudes of harmonic inputs to their resulting outputs.

Avoiding the need to fully identify the M and K matrices directly.

Foundational Theory: Primal vs. Dual Assembly

Defining the Connection (Allen et al. Taxonomy)

Substructuring methodologies are fundamentally categorised by how they enforce connection.

Primal Assembly

is the standard FEA approach (e.g., Craig-Bampton) where **unique global DOFs** are established for the entire assembly. Interface compatibility is satisfied *a priori* using localization matrices, which allows interface forces to cancel out, simplifying the calculation of the assembled system impedance [1].

$$\mathbf{Z}_g = \mathbf{L}^T \mathbf{Z} \mathbf{L}$$

Dual Assembly

In contrast to primal assembly, the Dual Assembly method keeps substructure DOFs separate and satisfies interface equilibrium *a priori* using Lagrange multipliers λ to represent interface forces. The solver then enforces **displacement compatibility** through a constraint equation.

$$\mathbf{B} \mathbf{u} = \mathbf{0}$$

This method is preferred for **experimental data (FBS)** as it handles noise and imperfect connectivity more robustly [1].

The dual approach is often preferred when working with experimental data, as it is more robust against measurement noise and imperfect connectivity.

Classification of Numerical and Experimental Approaches

While substructuring originated as a computational tool to manage large Finite Element models, its application has evolved significantly

The Craig-Bampton Method

The Numerical Standard (Fixed Interface)

The Craig-Bampton method is a dominant primal assembly technique used in industry. It represents the component's motion using a basis defined by two key sets of modes:

- **Constraint Modes** (ψ_{ib}), which are static shapes from unit displacements at the boundary
- **Fixed-Interface Normal Modes** (ϕ_{ik}), which are the natural vibration modes calculated with the boundaries rigidly clamped



Why It's Popular

because it **preserves the physical interface DOFs** for easy coupling and includes **static constraint modes** to ensure the reduced model maintains **accuracy and stiffness** at the boundary.

The total motion transformation is given by:

$$\{u\} = [\Phi_{ik} \quad \Psi_{ib}] \begin{Bmatrix} q \\ u_b \end{Bmatrix}$$

- **Construct Transformation Matrix** , Assemble the matrix that transforms physical coordinates into Craig-Bampton coordinates

$$T_{CB} = \begin{bmatrix} \Phi_{ik} & \Psi_{ib} \\ 0 & I_{bb} \end{bmatrix}$$

- **Reduce Matrices** by Pre-multiplying and post-multiplying the original mass and stiffness matrices by TCB to generate the reduced block-diagonal system matrices.

$$\mathbf{M}_{\text{reduced}} = \mathbf{T}_{CB}^T \mathbf{M} \mathbf{T}_{CB}$$

$$\mathbf{K}_{\text{reduced}} = \mathbf{T}_{CB}^T \mathbf{K} \mathbf{T}_{CB}$$

MacNeal's Method

The Hybrid Solution (Free Interface)

Whilst fixed-interface testing is difficult, free-interface testing is easier but can lead to significant accuracy loss when high-frequency modes are truncated. MacNeal (1971) solved this by introducing essential residual terms to maintain static accuracy.



Residual Flexibility

He defined Residual Flexibility ($Z^r(r)$) to mathematically correct for the static flexibility contributed by the high-frequency modes that were omitted.



Residual Mass

Simultaneously, he introduced the Residual Mass ($M^r(r)$) term to correct for inertial effects when modes are restrained.



Hybrid Accuracy

This hybrid approach significantly improves model accuracy, particularly in crucial fields like aeroelastic analysis.

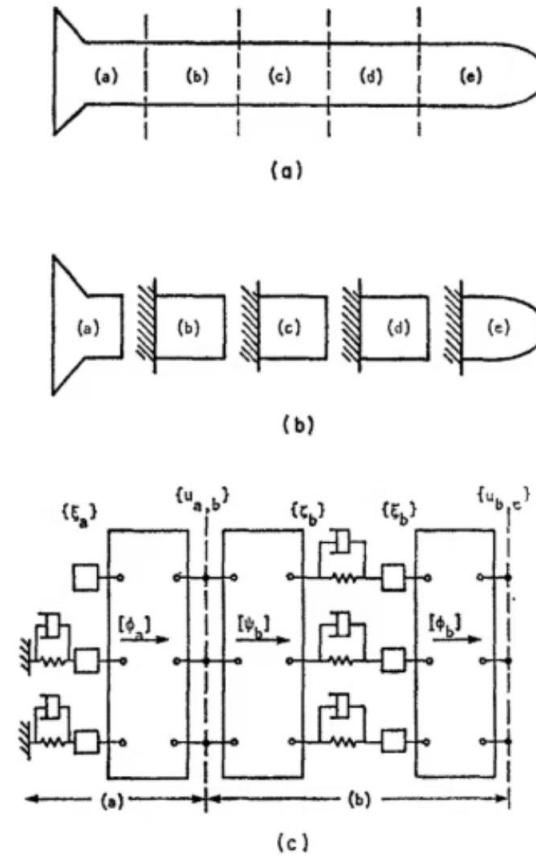


Figure 5: Dynamic partitioning of a missile structure

for hybrid component mode synthesis [3].

MacNeal demonstrated that including these residuals significantly improves accuracy in aeroelastic analyses, where control surface flexibilities are critical.

Modern Experimental ROM: The Transmission Simulator

Bridging the Gap Between Test and Theory

The inherent challenge of structural testing lies in fixture compliance, which corrupts the measurement of "fixed" boundaries.

The Transmission Simulator (TS) method addresses this by using a component with known dynamics (the simulator) attached to the test article.

By mathematically subtracting the known simulator dynamics from the coupled test results, the method derives an accurate model of the component that is fully decoupled from the fixture's influence.

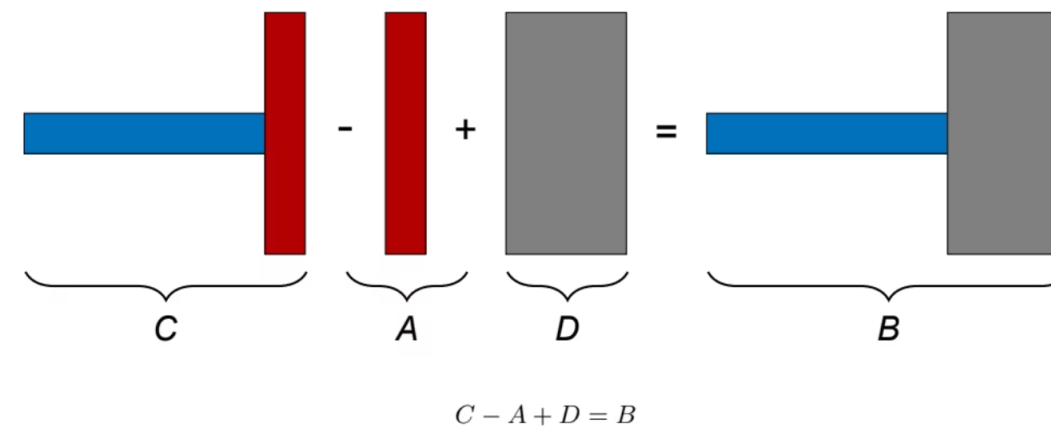


Figure 6: Diagram of the Transmission Simulator method illustrating the subtraction principle

This derived model effectively mimics the fixed-interface representation required for synthesis, successfully transferring experimental data into a format suitable for ROM synthesis.

The Transmission Simulator method operates on a subtraction principle:



1

Mass Loading:

The TS fixture is bolted to the component. Its mass and stiffness "exercise" the boundary degrees of freedom, ensuring that the experimental modes capture the relevant local dynamics at the connection points, which might be missed in a purely free-free test [4].

2

Mathematical Decoupling:

Since the TS is a known structure, its mass and stiffness can be mathematically subtracted from the coupled experimental results. This leaves a derived model of the component that effectively mimics the fixed-interface representation required for synthesis [4].

Comparative ROM Performance

Validation on Complex Assemblies

A key validation study, analysing a complex "Cylinder-Plate-Beam" assembly, confirmed the effectiveness of modern ROM techniques.



1

Accuracy

They achieved a very small frequency error (under 2%) compared to the true structural behavior, significantly outperforming older methods that ignored boundary dynamics [4].



Matrix Conditioning

If the condition number of these matrices exceeds 1013, the mathematical subtraction becomes unstable, often resulting in non-physical artifacts such as negative mass or imaginary frequencies [4].



Convergence Rates

To improve the convergence of the Transmission Simulator (TS) method, we must include the TS's **first elastic (bending) mode**, not just the rigid-body modes.

This provides a better mathematical basis for capturing how the interface moves, resulting in a **faster and more accurate solution** for the final system.

Implementation and Verification

Recovering Critical Physical Data

FEA solvers require specific LTMs to recover different aspects of the component's physical behavior:

Interface Force LTM:

To recover the net forces and moments acting at the structural boundary. This is critical for the design and sizing of bolted joints. It sums the inertial and stiffness contributions using the reduced C-B matrices.

$$\{F_{I/F}\} = \begin{bmatrix} M_{bb} & M_{bq} \\ 0 & K_{bb} \end{bmatrix} \begin{bmatrix} \ddot{u}_b \\ \ddot{q} \\ u_b \end{bmatrix}$$

Net CG Acceleration LTM:

The Net CG Acceleration LTM is a "virtual sensor" that captures the component's **net rigid-body acceleration** at the Center of Gravity (CG) by using the boundary mass (M_{bb}) and a rigid-body transformation.

The PHIZ LTM

uses the "modal acceleration" approach to calculate displacements ($\mathbf{X}_p$) from accelerations ($\ddot{\mathbf{X}}_p$ and $\ddot{\mathbf{q}}$).

This provides **significantly higher fidelity** for recovering internal stresses because it **implicitly accounts for the static flexibility** of the truncated high-frequency modes.

Verification Checks for ROM Fidelity

Validating the Reduced Model

Rigorous validity checks must be performed before integrating any substructure to ensure the reduction process has not corrupted the physical properties. Two primary checks are mandatory:

Free-Free Mode Check:

We calculate the free-free eigenmodes of the reduced Craig-Bampton model and compare them against the free-free modes of the original, full-fidelity model.

A close match confirms that the mass and stiffness properties have been correctly condensed and that the boundary conditions were handled properly during reduction.

Gravity (Unit Acceleration) Check:

We apply a uniform unit acceleration field (1g) rigidly to the boundary DOFs. LTMs are then used to calculate the resulting reaction forces.

These calculated forces and moments must **exactly match** the total weight and the correct Center of Gravity moments of the component. This confirms the rigid-body transformations and mass distribution within the reduced model are geometrically consistent.

Case Study: Launch Vehicle Gust Analysis

Application to Multi-Stage Vehicle Dynamics

Launch vehicles face severe transient **wind gusts** during ascent. The simulation by Jayasidhan et al. demonstrates this application to a three-stage vehicle [6]. The analysis uses a specialized **GUST_SIMULATION** tool (FORTRAN 95) to manage two key complexities:

1. **Continuous Property Change:** The tool automatically generates unique FEMs for various time instants as propellant mass changes continuously [6].
2. **Load Modeling:** Gusts are modeled using a "1-cosine" profile, generating normal forces based on dynamic pressure q and the local angle of attack α [6].

Critical Structural Findings (Transonic Regime)

- **Mass Balancing Effect:** The heavy aft section (engines/first stage) acts as a fixed inertial pivot. Consequently, the lighter forward section (payload) exhibits **significantly larger displacements** to maintain center of mass equilibrium during bending [6].
- **Critical Load Cases:**
 1. **Maximum Displacement:** Found at **Mach 1.1** (10.216 mm at the satellite location).
 2. **Maximum Acceleration:** Occurred at **Mach 1.05**, showing different flight regimes stress different structural limits [6].

Conclusion:

The evolution of substructuring ensures robust analysis of complex systems.

The **Craig-Bampton method** is the numerical standard due to its computationally efficient, block-diagonal form [2].

The **MacNeal method** is crucial for static accuracy, using **residual terms** to fix stiffness errors caused by modal truncation [3].

Modern methods like the **Transmission Simulator** are necessary for experimental testing, but require strict attention to **matrix conditioning** to avoid numerical instability [4].

Ultimately, these high-fidelity models are **indispensable for flight safety** and accurate transient analysis.

References

- [1] M. Allen, D. Rixen, M. van der Seijs, P. Tiso, T. Abrahamsson, and R. Mayes, Substructuring in Engineering Dynamics: Emerging Numerical and Experimental Techniques. Springer, 2020.
- [2] R. R. Craig and M. C. C. Bampton, “Coupling of substructures for dynamic analyses,” AIAA Journal, vol. 6, no. 7, pp. 1313–1319, 1968.
- [3] R. H. MacNeal, “A hybrid method of component mode synthesis,” Computers & Structures, vol. 1, no. 4, pp. 581–601, 1971.
- [4] D. Roettgen, B. Seegar, W. C. Tai, S. Baek, T. Dossogne, M. Allen, R. Kuether, M. R. W. Brake, and R. Mayes, “A Comparison of Reduced Order Modeling Techniques Used in Dynamic Substructuring,” Sandia National Laboratories, SAND2015-9121C, 2015.
- [5] S. Gordon, “The Craig-Bampton Method,” FEMCI Presentation, Goddard Space Flight Center, May 6, 1999.
- [6] A. K. Jayasidhan, J. Rose K, and R. Neetha, “Dynamic Response of a Launch Vehicle to Wind Gust,” International Journal of Engineering Development and Research, vol. 3, no. 3, 2015.